Nicolás Kass

Software architect & data engineer

I build the operational systems small businesses actually run on — inventory, point of sale, invoicing, field tooling — and the data pipelines behind them. Backend, frontend, deployment and the operation itself, because for most of my career there was nobody else to hand it to. I came to software from field biology, which is where I learned that a number needs a defence.

La Plata, Argentina · remote-friendly   github.com/nicolaskass   LinkedIn   t3.com.ar

WHAT I HAVE PUBLISHED

Four public repositories documenting how my systems are built and why. No client source code — the reasoning: what I decided, what I rejected, and what each decision cost. **34 architecture decision records** between them.

Specialty Retail ERP

github.com/nicolaskass/specialty-retail-erp

A production ERP · 13 modules · 84 models · 229 routes · ~190k LOC

System architecture at real scale: modules as bounded contexts with an acyclic dependency graph, and a consolidation refactor that unified four transaction types and three payment systems over live financial data.

Django · DRF · MongoDB · React · Docker · Traefik

Geospatial Survey Toolkit

github.com/nicolaskass/geospatial-survey-toolkit

Field tooling for a land-surveying practice

Road-works quality control, survey-code validation and a raster-to-CAD vectoriser, delivered to surveyors on Windows laptops with no Python and sometimes no internet. The tool flags anomalies and refuses to correct them: field data is evidence.

Python · Shapely · Streamlit · ezdxf · potrace

Geospatial Data Processing

github.com/nicolaskass/geospatial-data-processing

LiDAR & photogrammetry pipeline · reference book, 67 chapters

Measurement discipline applied to terrain data: validation against check points never used as control, uncertainty propagated to the delivered volume as two honest bounds, and quality gates that refuse rather than return a plausible wrong answer.

LiDAR · Photogrammetry · PDAL · GDAL · rasterio · Quarto

Ecological Data Analysis

github.com/nicolaskass/ecological-data-analysis

Statistics behind four manuscripts · 4 published reproducibility packages

Bayesian capture–recapture implemented directly and cross-checked against an established reference, with prior sensitivity analysis and a simulation of my own estimator's bias. Every reported number is generated, never transcribed.

Bayesian inference · MCMC · NumPy · SciPy · Reproducibility

HOW I WORK

The parts that are not standard. Django and React do not distinguish anyone; these do.

I make drift impossible instead of promising to prevent it

A post-commit hook that flags documentation gone stale against the code. A checker that verifies every citation from code to manuscript still resolves. A generated file that a paper reads its numbers from, so text and tables cannot disagree. Three domains, three times the same conclusion: anything maintained by intention decays — bind it mechanically.

I meter what I spend

Every LLM call in production writes tokens, latency and estimated cost to a usage log priced from a maintained table. Cost per feature is a query, not a surprise at the end of the month. Inference is a purchase, and systems meter what they buy.

I design what happens when a dependency is gone

Optional integrations degrade instead of failing: the provider SDK is imported at the call site, every entry point returns a structured result rather than raising, and the business flow completes with one field empty. A third-party integration has sat broken in production for months, by design, with no user-visible failure.

I know where automation should stop

In the surveying tools the software classifies anomalies across three severity levels and corrects none of them, because a miscoded point is sometimes a typo and sometimes the only real defect in the job. Automating a judgement is only safe when being wrong is cheap.

I write down what a decision cost

Every decision record has a section for its downsides, including the cases where I would choose differently for a different problem. A document that lists only benefits is marketing, and it makes everything around it less believable.

EXPERIENCE

Founder · Process & systems consultant

T³ — Three Tech Thinkers · La Plata, Argentina

2023 – present

- Operational diagnosis for small businesses: process mapping, quantification of time and money lost, and an intervention plan prioritised by impact.
- End-to-end project direction: scope, implementation sequence, stakeholder management, and evaluation against the objectives set in the diagnosis.
- Full-stack development of custom management systems — Python/Django and Node.js on the backend, React with Vite on the frontend, MongoDB and SQL, REST APIs and third-party integrations.
- LLM APIs integrated into production systems, with per-call cost metering and graceful degradation so an outage never blocks the business flow.
- Own infrastructure: Linux, Docker and Compose, Traefik and Nginx, VPS deployment with SSL, CI/CD on GitHub Actions.
- Process redesign and documentation under ISO 9001 criteria.

Head of operations & systems development

Digital Discos · La Plata, Argentina

2022 – 2026 · full responsibility from 2024

- Directed the complete operation of a 30-year-old family retail business: purchasing, sales, customer service, staff, stock and suppliers.
- Designed and built the business's management system: inventory built for 200,000+ *unique* catalogue items — distinct records with their own price and stock, not repeated units.
- Marketplace API integration for listing, price control and monitoring against the real market.
- Automated the order cycle end to end: intake with live catalogue, price and availability, and sale processing without manual intervention.
- The business went from selling only over the counter to operating nationally without growing its structure.

University lecturer & researcher

Universidad Nacional de La Plata (UNLP) · Argentina

2008 – present · formal appointment since 2010

- Research in natural sciences and undergraduate teaching. Design of studies with no prior art, data analysis, and communication of results.
- Entered through research scholarships, internships and teaching-assistant collaboration in 2008; formally appointed in both research and teaching from 2010.
- PhD candidate in Natural Sciences. Four manuscripts in review or preparation, each with a public reproducibility package.
- This is the direct origin of the method I apply in consulting: understand the system before intervening, measure against a defined objective, and adjust on evidence rather than intuition.

External process consultant · ISO 9001

Independent

2022

- Certified Internal Auditor, ISO 9001:2015 Quality Management Systems — Bureau Veritas Argentina, May 2022.
- Redesign of operational processes and procedures under ISO 9001, identification and quantification of unrecorded time and financial losses, and quality management system documentation.

Field technician · Ecosystem monitoring

Antarctic programme — [[ANT_ORG]]

[[ANT_YEAR]] · five months on station

- Ecosystem monitoring under formal sampling protocols: standardised procedures, instrument calibration, documented data custody, and quality requirements that leave no room for improvisation.
- Antarctic programmes operate to some of the strictest field-data standards there are, because a sample cannot be retaken and the season does not come back. This is where protocol discipline stopped being an abstraction for me — and it is the direct ancestor of how I treat measurement, validation and evidence in software.

Alumni · Project co-leader

Conservation Leadership Programme

2020 · 2022

- International training programme for conservation leaders: project planning, project management, stakeholder engagement and strategic communication. Global virtual cohort in 2020; in-person in South Africa in 2022, with LATAM and Africa cohorts.
- Future Conservationist Award as team co-leader.

TECHNICAL SKILLS

Backend	Python · Django · Django REST Framework · Node.js · Express · REST API design · JWT and OAuth · MongoDB / MongoEngine · SQL
Frontend	React · Vite · TypeScript · JavaScript · Tailwind · TanStack Query · HTML/CSS
Infrastructure	Linux · Docker & Compose · Traefik v3 · Nginx · VPS operation · Let's Encrypt · GitHub Actions · MCP servers
Geospatial	GDAL/OGR · PDAL · rasterio · Shapely · GeoPandas · LiDAR classification · photogrammetry (SfM/MVS) · DTM/DSM · DXF · GNSS RTK/PPK
Data & statistics	NumPy · SciPy · pandas · Bayesian inference and MCMC · capture–recapture models · non-parametric methods · uncertainty propagation · Matplotlib
Practice	Architecture decision records · modular monolith design · schema migration discipline · testing (pytest, Vitest, Playwright) · documentation as code (MkDocs, Quarto) · ISO 9001
AI tooling	Anthropic API in production · agent orchestration with version-controlled role definitions and per-role model-cost tiers · usage metering and cost attribution

EDUCATION & LANGUAGES

PhD candidate, Natural Sciences

Universidad Nacional de La Plata · conservation of Patagonian amphibians

In progress

Licenciado en Biología (Zoology) — equivalent to MSc

Universidad Nacional de La Plata

Certified Internal Auditor — ISO 9001:2015

Bureau Veritas Argentina · 2022

Languages

Spanish — native · English — professional working proficiency · German — B1 (certified)

Academic record

Publications, grants, teaching and conferences: ResearchGate. Full academic CV available on request.

Software

Programming since childhood and taught throughout school; self-taught since, working on Linux exclusively since around 2010. No formal computing qualification — the repositories above are the argument.

Every figure quoted here is a real measurement taken from the projects — lines, modules, routes and records counted from the source, not estimated. Client identifiers and deployment hostnames are omitted throughout.

Last updated: September 2026 · Versión en español